

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 30-304-1
MEDIUM VOLTAGE SAFETY INSULATED FOOTWEAR“SHOES”	1/11/2022

EDMS 30-304-1

SPECIFICATION

FOR

MEDIUM VOLTAGE SAFETY INSULATED

FOOTWEAR“SHOES”

Issue: NOV.-2022/ Rev- 1

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1- Scope:

This specification defines the minimum technical requirements for an electrically insulated footwear designed for protection against electrical shocks during live work in installations with nominal voltage up to 22 kV AC .

2- Applicable Codes and Standards:

The electrical insulating footwear should comply with the following two standards:

EN ISO 20345: 2011: Personal protective equipment – Safety footwear.

EN 50321-1: 2018: Live working – Footwear for electrical protection – Insulating footwear and over boots.

EN ISO 20344 Personal protective equipment — Test methods for footwear

Any provision of this specification differs from those of the standards listed above, the provision of this specification shall apply.

3- Definition:

The electrically Insulated footwear is defined as:

A Footwear which protects the wearer against electrical shocks by preventing the passage of dangerous current through the body via the feet.

4- General:

According to EN ISO 20345:2011:

- Any safety footwear shall be classified in accordance with Table 1.
- The safety footwear shall comply the basic requirements of EN ISO 20345: 2011, Table 2.
- The safety footwear may also have some additional features or requirements such as those listed in Table 3.

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Table (1)

Classification of footwear according to EN ISO 20345:2011.

Classification	Description
Class II	All-rubber (i.e. entirely vulcanized) or all polymeric (i.e. entirely molded) footwear.

Table (2)

Some chosen basic requirements for safety footwear (for class II only) according to EN ISO 20345:

Requirements		Sub clause in 20345:2011	Class II
Design	Height of the upper	5.2.2	✓
	Seat region	5.2.3	✓
Whole footwear	Toe protection: General	5.3.2.1	✓
	Internal length of toecaps	5.3.2.2	✓
	Impact resistance	5.3.2.3	✓
	Compression resistance	5.3.2.4	✓
	Behavior of toecaps	5.3.2.5	✓
	Leak proofness	5.3.3	✓
	Specific ergonomic Features	5.3.4	✓
	Slip resistance	5.3.5	✓
Upper	Thickness	5.4.2	✓
	Tensile properties	5.4.4	✓
	Flexing resistance	5.4.5	✓
	Hydrolysis	5.4.8	✓
Outsole	Design	5.8.1	✓
	Tear strength	5.8.2	✓
	Abrasion resistance	5.8.3	✓
	Flexing resistance	5.8.4	✓
	Hydrolysis	5.8.5	✓

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Table (3)

– Some chosen additional requirements in the safety footwear according to EN ISO 20345:

Requirements		Clause	Class II	Symbol
Whole Footwear	Penetration Resistance	6.2.1	✓	P
	Electrically insulating footwear	6.2.2.3	✓	See EN-50321
Outsole	Resistance to hot contact	6.4.1	✓	HRO

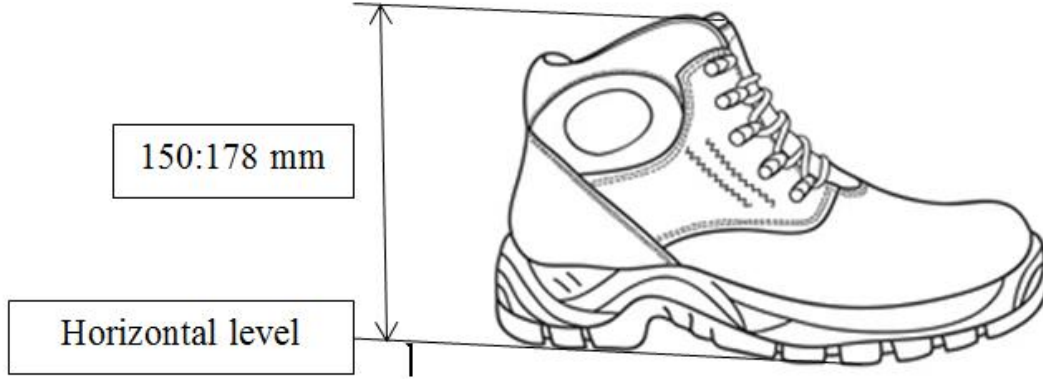
Note :

- Electrically insulating footwear : according to EN 50321 clause 4.4.2 symbol IEC standard 60417- 5216- 2001- 10- suitable for life working . (Double triangle for class 2)
- Resistance to hot contact (400 °C) according to EN 20344 clause 8.7.1.2

5- Design Criteria and Essential Features Required:

- The required safety footwear made from All-rubber material or all polymeric material.
- The required safety footwear is an electrical insulating footwear which must be classified as Class II [All-rubber (i.e. entirely vulcanized) or all polymeric (i.e. entirely moulded) footwear].
- The required safety footwear has an Electrical Class 0 (see electrical classification – Sub-clause (4.1), page 7 in EN 50321- 1: 2018) It is intended and needed for electrical protection during live working in installations with nominal voltage up to 17000 V AC or 25500 V DC.
- The required footwear design is: **B “Ankle boot”**, with height between (150:170 mm) from the horizontal level . See Figure 1, page 8 in EN ISO 20345:2011.”.

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- The required footwear must also fulfill all the basic requirements mentioned in the above given Table 2 in this specification.
- The required footwear must also fulfill all the additional safety requirements mentioned in Table 3 in this specification.
- The whole footwear must be resistant to penetration, Table 3.
- The outsole of the footwear must be resistance to hot contact, Table 3.

6- **Marking:**

- According to Clause 7 Markings, page 24 in EN ISO 20345: 2011., Each item of the safety footwear shall be clearly and permanently marked, by embossing or branding, with the following:
 - a) Size;
 - b) Manufacturer's identification mark;
 - c) Manufacturer's type designation;
 - d) Year and at least quarter of manufacture;
 - e) Reference to the International Standard, i.e. ISO 20345:2011;
 - f) Symbol(s) from Table 2 and Table 18 appropriate to the protection provided and/or, where applicable. (such as symbol “P” and symbol “HRO”).
 - g) For any additional marking on the footwear related to safety, the manufacturer shall provide evidence to support the claim and an explanation in the user notice.

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- According to Sub-clause 4.4 Markings, page 10 and 11 in EN 50321-1: 2018, Each item of the safety footwear shall be clearly and permanently marked, by embossing or branding, with the following additional marking:
 - a- symbol IEC60417-5216:2002-10-Suitable for live working; Double triangle. (double triangle for class 2)
 - b- number of EN 50321:201X standard immediately adjacent to the symbol;
 - c- electrical class; (class 1)
 - d- voltage current tested “AC” or “AC/DC” according to voltage testing applied.
- In addition, each unit of electrical insulating footwear shall have a strip to note the date of first use, the date of examination or the date of each periodic inspection. This shall be located near the IEC Symbol.
- Marking shall be on the external surface of the electrical insulating footwear and shall be clearly visible and does not need magnification.
- The marking shall be durable and shall remain always visible.

7- Packaging:

The packaging type suitable for transportation of the electrical insulating footwear shall be defined by the manufacturer.

Each pair of electrical insulating footwear shall be packed in an individual container or package.

The outside of packaging shall be marked with the name of the manufacturer or supplier, classification, size and design.

8- Information to be supplied by the manufacturer:

According to Clause 4.6 in EN 50321-1:2018, pages 11 and 12:

Each pair of electrical insulating footwear shall be accompanied by the instruction for use, which contain the information necessary for use, maintenance and potential risk of limited effectiveness of electrical insulation according to condition of use (i.e. mechanical or chemical aggression). The information supplied by the manufacturer shall conform to EN ISO 20345: 2011, 8.1 or EN ISO 20346: 2014, 8.1 or EN ISO 20347: 2012, 8.1.

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The instructions for use shall contain:

- Explanation of the symbol “double triangle”.
- Storage;
- Examination before use;
- Precaution of use;
- Periodic inspection

Beside the above instruction and according to Clause 8 in EN ISO 20345:2011, pages 25 & 26:

The safety footwear shall be supplied to the customer with information written at least in the official language(s) of the country of destination. All information shall be unambiguous and shall include the following:

- Name and full address of the manufacturer and/or his authorized representative.
- Notified body involved in type examination; for category III products, the notified body involved with Article 11 of EU Directive 89/686/EEC.
- Reference to the International Standard, i.e. ISO 20345: 2011.
- Explanation of any pictograms, markings and levels of performance.
- Basic explanation of the tests that have been applied to the footwear, if applicable.
- Instructions for use:
 - Tests to be carried out by the wearer before use, if required;
 - Fitting and how to put and take off the footwear, if relevant.;
 - Application (basic information on possible uses and where detailed information is give, the source);
 - Limitations of use (e.g. temperature range);
 - Instructions for storage and maintenance with maximum periods between maintenance checks (if important, drying procedures to be defined);
 - Instructions for cleaning and/or decontamination;
 - Obsolescence deadline or period of obsolescence;
 - If appropriate, warnings against problems, likely to be encountered (modifications can invalidate the type approval, e.g. orthopaedic footwear);

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- 9) If helpful, additional illustrations, part numbers, etc.
- g) Reference to accessories and spare parts, if relevant.
- h) Type of packaging suitable for transport, if relevant.

The outsole of the footwear must have resistance to hot contact and will be tested in accordance with ISO 20344:2011, 8.7, as rubber and polymeric outsoles shall not melt and shall not develop any cracks.

9- Non Electrical Tests:

Egyptian Organization For Standardization & Quality (EOS)

1. Impact resistance test
2. Compression resistance of safety footwear at 15 KN
3. water resistance test
4. Leather/rubber material test
5. Dry Antistatic Test
6. Outsole tear resistance test
7. Uppertear resistance test
8. Screw penetration resistance test

Chemistry Administration

- Thermal insulation test
- Lining treatment against bacteria

10-Electrical Tests:

Electrical insulating footwear shall pass an AC proof voltage test and withstand voltage test according to its electrical classification which is Class 2 in our case.

All the details required for applying this test are mentioned in EN 50321-1: 2018, pages 12 ~ 15.

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Table -3: Proof test voltage, proof test current and withstand test voltage for footwear

Class of footwear	AC Proof test voltage kV r.m.s.	AC Proof test current mA r.m.s	AC Withstand test voltage kV r.m.s	DC Proof voltage average kV
		Design (EN ISO 20345/EN ISO 20347)		
		B		
1	11	2.5	17	30
2	22	2.5	30	50

Where:

AC proof test voltage: Specified voltage that is applied to an electrical insulating footwear for the time defined under specified conditions to assure the electrical strength of the insulation is above a specified value.

Each unit of the electrical insulating footwear class 1 shall be submitted to an AC proof voltage test (17 kV AC) according to Table 3. The test period shall be 3 min, considered to start at the instant the specified proof test voltage is reached. The AC voltage shall be initially applied at a low value and will provide a constant rate of rise and decrease to the test voltage of approximately 17 kV/s either continuously or at the end of the period.

Each unit of the electrical insulating footwear class 2 shall be submitted to an AC proof voltage test (22 kV AC) according to Table 3. The test period shall be 3 min, considered to start at the instant the specified proof test voltage is reached. The AC voltage shall be initially applied at a low value and will provide a constant rate of rise and decrease to the test voltage of approximately 30 kV/s either continuously or at the end of the period.

The AC proof test is considered as passed if the specified test voltage is reached and maintained for the specified duration and the proof test current does not exceed the relevant value specified in Table 3 above.

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AC Withstand Test: Specified voltage that an electrical insulating footwear withstand without disruptive discharge or other electric failure when voltage is applied under specified conditions.

DC proof test voltage:

Each unit of the electrical insulating footwear class 1 shall be submitted to an DC test voltage (30 kV DC) according to Table 3. The test period shall be 1 min, considered to start at the instant the specified proof test voltage is reached. The AC voltage shall be initially applied at a low value and will provide a constant rate of rise and decrease to the test voltage of approximately 30 kV/s until the specified voltage level is reached or failure occurs. The DC proof test is considered passed if no failure occurs during the test period.

Each unit of the electrical insulating footwear class 2 shall be submitted to an DC test voltage (50 kV DC) according to Table 3. The test period shall be 1 min, considered to start at the instant the specified proof test voltage is reached. The AC voltage shall be initially applied at a low value and will provide a constant rate of rise and decrease to the test voltage of approximately 50 kV/s until the specified voltage level is reached or failure occurs. The DC proof test is considered passed if no failure occurs during the test period.

Three pairs (left and right) of electrical insulating footwear are to be tested, they shall meet the test requirements, including one pair from the smallest size, one pair from the largest and one pair of the middle size. After electrical testing, the electrical insulating footwear shall not be used. These tests are considered as destructive.

11-Test Report:

According to 5.2.5 page 17 in EN 50321-1: 2018, the test report issued for the footwear tested shall include the following:

- Class Test voltage;
- Identification and describe of the product;
- Reference of the standard;
- Conformity to the environmental conditions of testing;
- Date of the testing;
- Size and foot (left and right);

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- Ac proof test current measured.
- Qualification of testing laboratory.

12-General Requirements:

--EDC has the right to any non-electrical test in an approved laboratory that it deems appropriate to ensure the quality of the product and its achievement of the required specifications, at the supplier's expense.

Electrical tests will be carried out at the **High Voltage Laboratory of the Egyptian Electricity Holding Company or Cairo University laboratories and priority for High Voltage Laboratory of the Egyptian Electricity Holding Company.**

The supplier must provide the following:

- The validity of the shoes must be clarified from the date of production or from the date of use.
- An original certificate of origin for the country in which the shoes were manufactured.
- Certificates from accredited laboratories for all tests passed by the product.
- --EDC has the right to replace inappropriate sizes within three months of supplying.